(ii) Let $G = (V, E)$ be a cubic graph. We assume first that G has a $\mathbb{Z}_2^2$-flow f, and define an edge colouring $E \to \mathbb{Z}_2^2 \setminus \{0\}$. As $a = -a$ for all $a \in \mathbb{Z}_2^2$, we have $f(\vec{e}) = f(\overleftarrow{e})$ for every $\vec{e} \in \vec{E}$; let us colour the edge e with this colour $f(\vec{e})$. Now if two edges with a common end v had the same colour, then these two values of f would sum to zero; by (F2), f would then assign zero to the third edge at v. As this contradicts the definition of f, our edge colouring is correct.

Conversely, since the three non-zero elements of $\mathbb{Z}_2^2$ sum to zero, every 3-edge-colouring $c\colon E \to \mathbb{Z}_2^2 \setminus \{0\}$ defines a $\mathbb{Z}_2^2$-flow on G by letting $f(\vec{e}) = f(\overleftarrow{e}) = c(e)$ for all $\vec{e} \in \vec{E}$. □

Corollary 6.4.6. *Every cubic 3-edge-colourable graph is bridgeless.* □

6.5 Flow-colouring duality

In this section we shall see a surprising connection between flows and colouring: every k-colouring of a plane multigraph gives rise to a k-flow on its dual, and vice versa. In this way, the investigation of k-flows on arbitrary graphs, not necessarily planar, appears as a natural generalization of the familiar map colouring problems in the plane.

Let $G = (V, E)$ and $G^* = (V^*, E^*)$ be dual plane multigraphs. (This implies that G and G^* are connected; see Chapter 4.6.) For simplicity, let us assume that G and G^* have neither bridges nor loops and are non-trivial. For edge sets $F \subseteq E$, let us write $G = (V, E)$ $G^* = (V^*, E^*)$

$$F^* := \{\, e^* \in E^* \mid e \in F \,\}.$$

F^*

Conversely, if a subset of E^* is given, we shall usually write it immediately in the form F^*, and thus let $F \subseteq E$ be defined implicitly via the bijection $e \mapsto e^*$.

Suppose we are given a circulation g on G^*: how can we employ the duality between G and G^* to derive from g some information about G? The most general property of all circulations is Proposition 6.1.1, which says that $g(X, \overline{X}) = 0$ for all $X \subseteq V^*$. By Proposition 4.6.1, the bonds $E^*(X, \overline{X})$ in G^* correspond precisely to the cycles in G. Thus if we take the composition f of the maps $e \mapsto e^*$ and g, and sum its values over the edges of a cycle in G, then this sum should again be zero. Our first aim is to formalize and prove this observation.

Of course, there is still a technical hitch: since g takes its arguments not in E^* but in $\vec{E^*}$, we cannot simply define f as above: we first have to refine the bijection $e \mapsto e^*$ into one from $\vec{E}$ to $\vec{E^*}$, i.e. assign to every $\vec{e} \in \vec{E}$ canonically one of the two orientations of e^*. This will be the purpose of our first lemma. After that, we shall show that f does indeed sum to zero along any cycle in G.

If $C = v_0 \dots v_{\ell-1} v_0$ is a cycle with edges $e_i = v_i v_{i+1}$ (and $v_\ell := v_0$), we shall call

$\vec{C}$

$$\vec{C} := \{\, (e_i, v_i, v_{i+1}) \mid i < \ell \,\}$$

cycle with orientation

a *cycle with orientation*. Note that this definition of $\vec{C}$ depends on the vertex enumeration chosen to denote C: every cycle has two orientations. Conversely, of course, C can be reconstructed from the set $\vec{C}$. In practice, we shall therefore speak about C freely even when, formally, only $\vec{C}$ has been defined.

Lemma 6.5.1. *There exists a bijection* $^*\colon \vec{e} \mapsto \vec{e}^{\,*}$ *from* $\vec{E}$ *to* $\overrightarrow{E^*}$ *with the following properties:*

(i) *The underlying edge of* $\vec{e}^{\,*}$ *is always* e^*, *i.e.* $\vec{e}^{\,*}$ *is one of the two orientations* $\overrightarrow{e^*}, \overleftarrow{e^*}$ *of* e^*;

(ii) *If* $C \subseteq G$ *is a cycle,* $F := E(C)$, *and if* $X \subseteq V^*$ *is such that* $F^* = E^*(X, \overline{X})$, *then there exists an orientation* $\vec{C}$ *of* C *with* $\{\, \vec{e}^{\,*} \mid \vec{e} \in \vec{C} \,\} = \overrightarrow{E^*}(X, \overline{X})$.

The proof of Lemma 6.5.1 is not entirely trivial: it is based on the so-called *orientability* of the plane, and we cannot give it here. Still, the assertion of the lemma is intuitively plausible. Indeed if we define for $e = vw$ and $e^* = xy$ the assignment $(e, v, w) \mapsto (e, v, w)^* \in \{(e^*, x, y), (e^*, y, x)\}$ simply by turning e and its ends clockwise onto e^* (Fig. 6.5.1), then the resulting map $\vec{e} \mapsto \vec{e}^{\,*}$ satisfies the two assertions of the lemma.

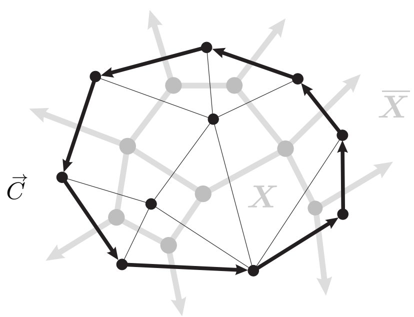

Fig. 6.5.1. Oriented cycle-cut duality

$\vec{e}^{\,*}$

f, g

Consider a fixed bijection $^*\colon \vec{e} \mapsto \vec{e}^{\,*}$ as provided by Lemma 6.5.1. Given an abelian group H, let $f\colon \vec{E} \to H$ and $g\colon \overrightarrow{E^*} \to H$ be two maps such that

$$f(\vec{e}) = g(\vec{e}^{\,*})$$

for all $\vec{e} \in \vec{E}$. For $\vec{F} \subseteq \vec{E}$, we set

$$f(\vec{F}) := \sum_{\vec{e} \in \vec{F}} f(\vec{e})\,.$$

$f(\vec{C})$ etc.

Lemma 6.5.2.

(i) *The map g satisfies (F1) if and only if f does.*

(ii) *The map g is a circulation on G^* if and only if f satisfies (F1) and $f(\vec{C}) = 0$ for every cycle $\vec{C}$ with orientation.*

(1.9.3)
(4.6.1)
(6.1.1)

Proof. Assertion (i) follows from Lemma 6.5.1 (i) and the fact that $\vec{e} \mapsto \vec{e}^{\,*}$ is bijective.

For the forward implication of (ii), let us assume that g is a circulation on G^*, and consider a cycle $C \subseteq G$ with some given orientation. Let $F := E(C)$. By Proposition 4.6.1, $F^* =: E^*(X, \overline{X})$ is a bond of G^*. By definition of f and g, Lemma 6.5.1 (ii) and Proposition 6.1.1 give

$$f(\vec{C}) = \sum_{\vec{e} \in \vec{C}} f(\vec{e}) = \sum_{\vec{d} \in \vec{E^*}(X,\overline{X})} g(\vec{d}) = g(X, \overline{X}) = 0$$

for one of the two orientations $\vec{C}$ of C. Then, by $f(\overleftarrow{C}) = -f(\vec{C})$, also the corresponding value for our given orientation of C must be zero.

For the backward implication it suffices by (i) to show that g satisfies (F2). Let $v \in V^*$ be given. By Lemma 1.9.3, the cut $E^*(v)$ is a disjoint union of bonds $D^* = E^*(X, \overline{X})$; let us name these so that always $v \in X$. Since every edge in these bonds is incident with v, we then have $\vec{E^*}(X, \overline{X}) \subseteq \vec{E^*}(v)$ also for the oriented edges.

By Proposition 4.6.1, each of the sets $D \subseteq E$ is the edge set of a cycle C in G, which by Lemma 6.5.1 (ii) has an orientation $\vec{C}$ such that

$$\{\, \vec{e}^{\,*} \mid \vec{e} \in \vec{C}\,\} = \vec{E^*}(X, \overline{X})\,.$$

Hence $g(X, \overline{X}) = f(\vec{C}) = 0$ by definition of f and g, giving

$$g(v, V^*) = \sum_X g(X, \overline{X}) = 0$$

as desired. □

With the help of Lemma 6.5.2, we can now prove our colouring-flow duality theorem for plane multigraphs. If $P = v_0 \dots v_\ell$ is a path with edges $e_i = v_i v_{i+1}$ ($i < \ell$), we set (depending on our vertex enumeration of P)

$$\vec{P} := \{\, (e_i, v_i, v_{i+1}) \mid i < \ell \,\}$$

$\vec{P}$

and call $\vec{P}$ a $v_0 \to v_\ell$ *path*. Again, P may be given implicitly by $\vec{P}$.

$v_0 \to v_\ell$ *path*

Theorem 6.5.3. (Tutte 1954)
For every dual pair G, G^* of plane multigraphs,

$$\chi(G) = \varphi(G^*)\,.$$

(1.5.5) V, E V^*, E^* *Proof.* Let $G =: (V, E)$ and $G^* =: (V^*, E^*)$. For $|G| \in \{1, 2\}$ the assertion is easily checked; we shall assume that $|G| \geqslant 3$, and apply induction on the number of bridges in G. If $e \in G$ is a bridge then e^* is a loop, and $G^* - e^*$ is a plane dual of G/e (why?). Hence, by the induction hypothesis,

$$\chi(G) = \chi(G/e) = \varphi(G^* - e^*) = \varphi(G^*)\,;$$

for the first and the last equality we use that, by $|G| \geqslant 3$, e is not the only edge of G.

So all that remains to be checked is the induction start: let us assume that G has no bridge. If G has a loop, then G^* has a bridge, and $\chi(G) = \infty = \varphi(G^*)$ by convention. So we may also assume that G has
k no loop. Then $\chi(G)$ is finite; we shall prove for given $k \geqslant 2$ that G is k-colourable if and only if G^* has a k-flow. As G – and hence G^* – has neither loops nor bridges, we may apply Lemmas 6.5.1 and 6.5.2 to G and G^*. Let $\vec{e} \mapsto \vec{e}^{\,*}$ be a bijection between $\vec{E}$ and $\vec{E^*}$ as in Lemma 6.5.1.

g f We first assume that G^* has a k-flow. Then G^* also has a $\mathbb{Z}_k$-flow g. As before, let $f\colon \vec{E} \to \mathbb{Z}_k$ be defined by $f(\vec{e}) := g(\vec{e}^{\,*})$. We shall use f to define a vertex colouring $c\colon V \to \mathbb{Z}_k$ of G.

Let T be a normal spanning tree of G, with root r, say. Put $c(r) := \overline{0}$. For every other vertex $v \in V$ let $c(v) := f(\vec{P})$, where $\vec{P}$ is the $r \to v$ path in T. To check that this is a proper colouring, consider an edge $e = vw \in E$. As T is normal, we may assume that $v < w$ in the tree-order of T. If e is an edge of T then $c(w) - c(v) = f(e, v, w)$ by definition of c, so $c(v) \neq c(w)$ since g (and hence f) is nowhere zero. If $e \notin T$, let $\vec{P}$ denote the $v \to w$ path in T. Then

$$c(w) - c(v) = f(\vec{P}) = -f(e, w, v) \neq \overline{0}$$

by Lemma 6.5.2 (ii).

c Conversely, we now assume that G has a k-colouring c. Let us define $f\colon \vec{E} \to \mathbb{Z}$ by

f $$f(e, v, w) := c(w) - c(v)\,,$$

g and $g\colon \vec{E^*} \to \mathbb{Z}$ by $g(\vec{e}^{\,*}) := f(\vec{e})$. Clearly, f satisfies (F1) and takes values in $\{\pm 1, \dots, \pm(k-1)\}$, so by Lemma 6.5.2 (i) the same holds for g. By definition of f, we further have $f(\vec{C}) = 0$ for every cycle $\vec{C}$ with orientation. By Lemma 6.5.2 (ii), therefore, g is a k-flow. □

6.6 Tutte's flow conjectures

How can we determine the flow number of a graph? Indeed, does every (bridgeless) graph have a flow number, a k-flow for some k? Can flow numbers, like chromatic numbers, become arbitrarily large? Can we characterize the graphs admitting a k-flow, for given k?

Of these four questions, we shall answer the second and third in this section: we prove that every bridgeless graph has a 6-flow. In particular, a graph has a flow number if and only if it has no bridge. The question asking for a characterization of the graphs with a k-flow remains interesting for $k = 3, 4, 5$. Partial answers are suggested by the following three conjectures of Tutte, who initiated algebraic flow theory on graphs.

The oldest and best known of the Tutte conjectures is his *5-flow conjecture*:

Five-Flow Conjecture. (Tutte 1954)
Every bridgeless multigraph has a 5-flow.

Which graphs have a 4-flow? By Proposition 6.4.4, the 4-edge-connected graphs are among them. The Petersen graph (Fig. 6.6.1), on the other hand, is an example of a bridgeless graph without a 4-flow: since it is cubic but not 3-edge-colourable, it cannot have a 4-flow by Proposition 6.4.5 (ii).

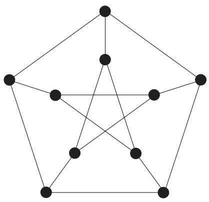

Fig. 6.6.1. The Petersen graph

Tutte's *4-flow conjecture* states that the Petersen graph must be present in every graph without a 4-flow:

Four-Flow Conjecture. (Tutte 1966)
Every bridgeless multigraph not containing the Petersen graph as a minor has a 4-flow.

By Proposition 1.7.3, we may replace the word 'minor' in the 4-flow conjecture by 'topological minor'.